Benchmarking a Polygon-Output Swin–Mask R-CNN Segmentation Pipeline Against Rectangle-Annotated Bengali HTR Baselines

Executive Summary Report — Three-Dataset Results

(Condensed from the full technical report; figures and inference-image galleries omitted, all numbers identical)

OCR Pipeline Benchmarking Report

September 2026

Abstract

We benchmark our Swin/Mask R-CNN pipeline (polygon output) against **BN-DRISHTI** [15] (YOLOv5, rectangle output) on three datasets using a literature-derived, format-agnostic protocol [1]. **Swin** wins decisively on synthetic data, its own training domain (word F1 0.915 vs. 0.684); **BN-DRISHTI** wins on real BN-HTRd, its training domain; and **Swin** still wins on BanglaWriting, ground neither system trained on (word F1 0.770 vs. 0.674). The result is dataset-conditional, not a single winner. **BN-DRISHTI**'s edge on strict mAP@75 is separately shown to be a rectangle-vs-polygon scoring artifact, not a real accuracy gap.

At-a-Glance: All Three Datasets

Dataset	Task/System	F1	mAP@50	mAP@75	mAP
Synthetic (`syn-hw`)	Line – **BN-DRISHTI**	0.776	0.508	**0.209**	0.239
***(Swin**'s home turf)*	Line – **Swin**	**0.893**	**0.804**	0.091	**0.272**
	Word – **BN-DRISHTI**	0.684	0.318	**0.172**	0.168
	Word – **Swin**	**0.915**	**0.688**	0.037	**0.214**
Real (BN-HTRd)	Line – **BN-DRISHTI**	**0.989**	**0.989**	**0.973**	**0.813**
***(BN-DRISHTI**'s home turf)*	Line – Swin	0.903	0.865	0.343	0.426
	Word – **BN-DRISHTI**	**0.685**	**0.432**	**0.420**	**0.362**
	Word – Swin	0.669	0.263	0.001	0.050
Neutral (BanglaWriting)	Word – **BN-DRISHTI**	0.674	**0.500**	**0.083**	**0.186**
(foreign to both)	Word – **Swin**	**0.770**	0.002	0.000	0.000

F1 is the headline metric in every table below (Wolf & Jolion split-merge matching [1], the ICDAR/DetEval standard); mAP@50/75/mAP are reported for completeness only and are known to be unreliable for word-level detection (small-box IoU sensitivity, plus one-to-one matching penalizing legitimate split/merge detections as false positives – footnote below). No single dataset row is "the" answer either: the two home-turf datasets each answer "which system is better *on its own* training distribution"; only the neutral BanglaWriting row approximates "which system generalizes better." Read together: **Swin** is the stronger raw detector in two of three conditions (including the unconfounded one), but loses decisively on mask-mode mAP@75 everywhere real, non-synthetic ground truth is used, for a geometric reason explained in Section 3.

1 Introduction and the Training-Domain Confound

Our pipeline (Swin Transformer backbone, hierarchical Mask R-CNN, page→line →word→grapheme cascade) emits a 50-point directional polygon per instance, closely tracing the true slant and curvature of handwritten Bengali text. **BN-DRISHTI** [15] and the BN-HTRd dataset it extends [14] instead represent every line/word as an axis-aligned rectangle. A naive rectangle-vs-rectangle IoU cannot score our polygon predictions without either erasing their shape information (bbox reduction) or inventing an unprecedented ad hoc comparison, so Section 2 adopts a literature-grounded, format-agnostic protocol.

The training-domain confound. **Swin** was trained *exclusively* on the synthetic `syn-hw` corpus; **BN-DRISHTI** was trained *exclusively* on the real BN-HTRd corpus. Neither system has ever seen the other's training distribution. This means a `syn-hw` benchmark is simultaneously an in-domain test for **Swin** and an

out-of-domain generalization test for **BN-DRISHTI**, and the real-BN-HTRd benchmark is exactly reversed – neither alone is a neutral referee on "which detector is generally better." We therefore report *three* datasets side by side: the two home-turf corpora, plus BanglaWriting [16], a third, independent corpus neither system has ever trained on, added specifically to settle the generalization question the first two cannot.

2 Literature Review and Adopted Protocol

We searched arXiv, IEEE Xplore, Springer Link, ACM DL, CVF Open Access, Semantic Scholar, DBLP, and the Robust Reading Competition portal for methodology reconciling polygon/quadrilateral/rectangle detection geometries under one evaluation protocol. Twenty-six sources were retained. Two mechanisms recur: (1) Wolf & Jolion's object count/area *split/merge* matching [1], the basis of "DetEval" used in early ICDAR Robust Reading Competitions [2, 3, 4], which credits a detection against several ground-truth regions once area-coverage constraints are jointly satisfied ($\gtrsim$0.4 per-region, $\gtrsim$0.8 total), directly addressing line/word granularity mismatches between cascaded models; and (2) rasterizing both sides to a shared pixel-mask canvas and scoring pixel IoU, exactly as COCO's `iou_type="segm"` [9] and DIVA-HisDB [10] do, sidestepping the polygon/rectangle format mismatch entirely. Total-Text [5, 6] independently establishes polygon-native ground truth for curved scene text, and Tightness-Aware IoU [7] warns that a single IoU threshold can conflate localization quality with shape-format agreement – a caution central to Section 3. Document layout analysis (PubLayNet [11], DocLayNet [12], FUNSD [13]) and the ICDAR Robust Reading lineage converge on P/R/F1 at IoU$\geq$0.5 as the primary bar, with COCO-style mAP@50/75/mAP as secondary, stricter numbers – a convention we adopt. Critically, *no* Bengali/Indic HTR paper surveyed (BN-HTRd, **BN-DRISHTI**, BanglaWriting, PHDIndic_11 [17]) defines a protocol for comparing a rectangle-output system against a polygon-output one; this report's methodology appears to be a first for the Bengali OCR space.

Adopted protocol: (i) implement Wolf & Jolion split/merge matching [1] underneath a strict greedy IoU$\geq$0.5 matcher and treat the resulting P/R/F1 as this report's **headline metric**, so line/word granularity mismatches do not silently inflate false positives/negatives; (ii) rasterize every prediction/ground-truth region onto a shared downscaled canvas (768 px longest side) and score COCO-style mask mAP, so **Swin**'s polygons are never penalized merely for not being rectangles, reported alongside for completeness; (iii) additionally report vector box-IoU mAP as a sanity check. For a pure rectangle-vs-rectangle case (**BN-DRISHTI** vs. ground truth), box-mode and mask-mode agree to within 1–2%, confirming the rasterization pipeline introduces no bias before it is trusted for the genuinely mismatched polygon case.

Why F1 leads and mAP follows.[1] COCO mAP was built for generic object detection, not small, densely-packed word boxes: IoU is a relative-area measure, so a $\sim$30–60 px word box swings sharply in IoU from sub-visual pixel misalignment, and strict one-to-one matching scores every extra overlapping detection as a flat false positive – collapsing precision-driven AP whenever a system legitimately splits one word into several correct sub-boxes, exactly the case split/merge matching handles correctly instead (Table 1 above: switching **BN-DRISHTI** from strict to split-merge matching raises word F1 from 0.487 to 0.684 with *no change to the underlying predictions*). CLEval's character-level scoring [8] and Tightness-Aware IoU [7] are further literature metrics purpose-built for this exact setting; both are computed for BanglaWriting (Section 6) and pending for `syn-hw`/BN-HTRd, with fuller detail in the companion full report.

3 Part I: Synthetic Corpus (`syn-hw`) Results

`syn-hw` is a synthetic handwritten Bengali page corpus (60,454 pages total, 6,045-page held-out validation split used here; 118,103 lines / 520,012 words in that split), each page mixing several distinct synthetic handwriting fonts. Ground truth is a bounding-box reduction of a native, pixel-precise 50-point ink-tracing polygon – computed specifically to match BN-HTRd/**BN-DRISHTI**'s own rectangle convention, not because finer ground truth is unavailable. This is precisely why the rectangle ground truth contains the "empty margin" quantified in Section 3.

Split-merge matching materially changes **BN-DRISHTI**'s own scores versus strict one-to-one matching (word F1 0.487$\rightarrow$0.684, line F1 0.690$\rightarrow$0.776), confirming granularity mismatches are common; mAP is unaffected since `MeanAveragePrecision` performs its own internal one-to-many ranking. The full three-way comparison (both systems vs. ground truth, split-merge matching, mask-mode IoU, 6,045 pages, 2.53 h on one RTX 5090):

[1] Full derivation and a consolidated mAP reference table: Appendix A of the companion full technical report (`report/benchmark_report.tex`).

Table 1: Full `syn-hw` validation split. F1 = headline metric; mAP reported for completeness only.

Task	System	P	R	F1	mAP@50	mAP@75	mAP
Line	**BN-DRISHTI**	0.897	0.684	0.776	0.508	**0.209**	0.239
Line	**Swin**	0.858	**0.931**	**0.893**	**0.804**	0.091	**0.272**
Word	**BN-DRISHTI**	0.546	0.916	0.684	0.318	**0.172**	0.168
Word	**Swin**	0.858	**0.979**	**0.915**	**0.688**	0.037	**0.214**

Swin wins F1 by 11.7/23.1 points and mAP@50 by 29.6/37.0 points (line/word), driven by much higher recall at comparable precision. The one metric **BN-DRISHTI** wins is mAP@75, investigated next.

4 Why Is BN-DRISHTI's mAP@75 Higher? (Investigation and Proof)

A dedicated multi-stage investigation ruled out artifact explanations before accepting a geometric one. **Resolution is not the cause**: mean matched-pair IoU is flat across a $\sim 8\times$ canvas-resolution sweep (384 px to native 2480×3510) for both systems, including at zero downscale. **Rasterization is not the cause**: `cv2.fillPoly`'s rasterized pixel count agrees with the exact shoelace-formula polygon area to within 1% (mean ratio 0.991, $n = 32$, zero malformed polygons). **Box-mode reduction flips the ordering entirely in Swin's favor** (reducing its polygon to its own bounding box turns the comparison into pure detection confidence, where **Swin** is simply stronger) – and a real handwritten line's tight **Swin** polygon covers on average only 73.4% of its own enclosing rectangle (55.9–96.4%, $n = 32$).

To make the mechanism fully concrete and independent of either model's own geometry, we Otsu-thresholded [20] the *source page images* inside every matched ground-truth box (60-page sample) to separate true ink pixels I from background/margin M. For matched prediction P: *ink recall* $= |P \cap I|/|I|$ and *background coverage* $= |P \cap M|/|M|$ – a purely pixel-intensity measurement using neither model's prediction as ground truth.

Table 2: Independent pixel-level ink/margin decomposition (60-page sample).

Task	System	Ink recall	Bg. coverage	n
Line	**BN-DRISHTI**	0.958	**0.857**	1,180
Line	**Swin**	0.933	0.670	1,177
Word	**BN-DRISHTI**	0.988	**0.979**	6,363
Word	**Swin**	0.975	0.576	6,489

Both systems recall real ink at a comparable, high rate (0.933–0.988): neither is meaningfully "missing" text. The gap is entirely in background coverage – **BN-DRISHTI**'s rectangle claims 85.7%/97.9% of each box's empty margin vs. **Swin**'s 67.0%/57.6% – the exact per-pixel mechanism behind the mAP@75 gap. A further check: only 8.2%/20.9% (line/word) of a typical ground-truth box is actually ink, so a hypothetically perfect ink-tight detector is structurally capped near IoU $\approx$0.08 for lines – strikingly close to **Swin**'s observed line mAP@75 of 0.091. **Verdict: mAP@75 under rectangle-only ground truth measures how rectangle-shaped a prediction is, not how accurate it is**; mAP@50, split-merge P/R/F1, and box-mode's own ordering are the metrics we recommend trusting, and **Swin** wins clearly on all of them, for the right reasons.

5 Part II: Real Corpus (BN-HTRd) Results

BN-DRISHTI's own paper reports headline numbers on real, photographed handwriting, so we locate the actual extended BN-HTRd (v4.0) dataset already present locally (`BN_HTR/`, 3.0 GB) and run the identical protocol on it. **Reading [15] directly**: its headline 99.97%(line)/98%(word) F-scores are computed by *manual visual comparison* ("Qualitative" evaluation), not an automated IoU protocol; its own automated ("Quantitative") evaluation scores only 94.09%(line)/16.0%(word, hurt by a disclosed skew-correction coordinate bug), and the paper candidly states this word evaluation "does not do justice" to the system. Neither of **BN-DRISHTI**'s own two numbers is therefore directly comparable to our fully automated protocol without this context.

The corpus contains 786 real pages/150 writers (matching the paper's reported 786 pages/14,383 lines); we use an existing writer-disjoint split whose test set (84 pages, 1,444 line boxes) is comparable in scale to the

paper's own 75-image test set. Word-level ground truth is only annotated per line crop, so we reconstruct 10,238 page-level word boxes by offsetting each line crop's word boxes by its line box's own origin (verified exact via crop-dimension matching).

Table 3: Real BN-HTRd, 84-page test split, identical protocol. F1 = headline metric; mAP reported for completeness only.

Task	System	P	R	F1	mAP@50	mAP@75	mAP
Line	**BN-DRISHTI**	0.978	**0.999**	**0.989**	**0.989**	**0.973**	**0.813**
Line	Swin	0.877	0.930	0.903	0.865	0.343	0.426
Word	**BN-DRISHTI**	0.524	**0.988**	**0.685**	**0.432**	**0.420**	**0.362**
Word	Swin	**0.828**	0.561	0.669	0.263	0.001	0.050

BN-DRISHTI wins decisively on lines and edges ahead on words. **Swin** keeps a real strength (word precision 0.828 vs. 0.524) but its recall collapses to 0.561 (vs. 0.979 on synthetic data) – it simply misses much real handwritten text. Our automated reproduction gives **BN-DRISHTI** a line F1 of 98.87% (between the paper's own 94.09% automated and 99.97% manual figures) and word F1 of 68.52% (far above the paper's flawed 16.0% automated figure but below its manual 98%) – evidence our reproduction is methodologically sound. This is a textbook synthetic-to-real domain gap: **Swin** was trained only on synthetic renders and never saw a real photographed page, while **BN-DRISHTI** was trained directly on this real corpus; it does not overturn Part I, it contextualizes it.

6 Part III: Neutral Corpus (BanglaWriting) Results

Both results above are confounded by training data. To answer "which detector generalizes better" directly, we benchmark on BanglaWriting [16]: 260 single-page documents, 260 different writers, 52 scanner-digitized + 208 smartphone-photographed under real lighting, 21,234 manually-annotated word boxes – **a corpus neither system has ever trained on**. BanglaWriting carries no official line-level ground truth (only words), and rather than fabricate line ground truth by heuristically clustering words – our own derived ground truth, not an independently verifiable one – we score the word task only, across all 260 pages.

Table 4: Word task, BanglaWriting (260 pages, neutral for both systems). F1 = headline metric; mAP reported for completeness only and especially unreliable here (Section 5).

System	P	R	F1	mAP@50	mAP@75	mAP
BN-DRISHTI	0.530	**0.924**	0.674	**0.500**	**0.083**	**0.186**
Swin	**0.790**	0.752	**0.770**	0.002	0.000	0.000

On data neither system has trained on, Swin still wins word-task F1 (0.770 vs. 0.674), driven by much higher precision at somewhat lower recall than **BN-DRISHTI** – a real, if narrower, win than its 59-point synthetic-data margin, not a tie and not a reversal. mAP again collapses for **Swin** in mask mode (0.002 mAP@50) while its box-mode mAP@50 stays respectable (0.353, close to **BN-DRISHTI**'s 0.500) – confirmed via a direct, pipeline-independent rasterized IoU check (172 matched instances): **Swin**'s mean polygon-vs-rectangle mask IoU is only 0.32–0.34 (0–1.2% clearing 0.5) while its box-mode IoU averages 0.51–0.57 (56–73% clearing 0.5). This is the same mechanism as Section 3, *more pronounced* here because BanglaWriting's human-drawn boxes carry more margin than `syn-hw`'s ink-derived ones – independent confirmation the mAP@75 effect is dataset-general, not specific to `syn-hw`'s ground-truth construction.

Literature cross-check (CLEval).[2]

[2]Run via the official, unmodified-algorithm upstream implementation [8]; a small third-party bug (a missing attribute in CLEval's polygon box type) had to be patched to run correctly on our polygon predictions – an upstream implementation bug, not an adjustment to our data. TIoU [7] was also run on BanglaWriting but is not reported as a system comparison here: it gates its continuous tightness score behind a hard IoU≥0.5 base match, and **Swin**'s raw ink-tracing polygon mostly fails that gate against BanglaWriting's box-only ground truth for the same geometric reason mask-mode mAP@50 collapsed above, making a polygon-vs-box TIoU comparison structurally unfair here rather than informative; a fair TIoU comparison is possible on `syn-hw`, which has real polygon ground truth, once that run completes.

Table 5: CLEval (character-level) cross-check, word task, BanglaWriting.

System	P	R	F1
BN-DRISHTI	0.544	**0.774**	0.639
Swin	**0.852**	0.752	**0.799**

CLEval's character-level F1 (0.639 **BN-DRISHTI**, 0.799 Swin) tracks the split-merge headline F1 closely for both systems and preserves the same ranking – independent confirmation, from an officially-maintained protocol with no dependence on this project's own matching code, that the headline numbers reflect real detection quality. `syn-hw`/BN-HTRd CLEval runs, and `syn-hw`'s polygon-vs-polygon TIoU run, are pending.

Resolving the confound: **Swin** wins decisively on its own synthetic training domain, **BN-DRISHTI** and **Swin** are near parity on BN-HTRd (**BN-DRISHTI**'s home turf), and on the one corpus foreign to *both*, **Swin** wins again by a smaller but real margin. The most defensible summary: **Swin** is the stronger raw word detector by F1 in two of three conditions, including the unconfounded one, but this advantage narrows away from its own training distribution and never extends to mask-mode mAP on real, non-synthetic ground truth, where the rectangle-vs-polygon geometric penalty dominates regardless of dataset.

7 Discussion

(0) Neither home-turf dataset alone is a neutral referee – Part III exists precisely to avoid averaging two biased numbers. **(1)** Matching strategy materially changes **BN-DRISHTI**'s own reported scores (word F1 understated by ∼20 points under strict matching); its own paper's headline 99.97%/98% figures are themselves manually adjudicated, not automated, complicating any comparison to automated protocols including ours. **(2)** The mAP@75 result is a caution against a single headline metric across systems with different native output geometries, as Tightness-Aware IoU [7] argues generally. **(3)** The real-BN-HTRd ranking reversal is a genuine, deployment-relevant synthetic-to-real domain gap for **Swin**, not yet closed by fine-tuning. **(4)** BanglaWriting shows this gap is tied to BN-HTRd's specific capture conditions (or the line task specifically), not broad evidence **Swin** is worse at handwriting detection once training-domain advantage is removed – on a third, differently-captured real corpus it is not. **(5)** The same neutral corpus reproduces the mAP@75 artifact independently and *more severely*, confirming it as a general property of comparing polygon output to rectangle ground truth.

8 Conclusion

On synthetic `syn-hw` data (our training domain), **Swin** beats **BN-DRISHTI** decisively on every fair metric. On real BN-HTRd (**BN-DRISHTI**'s training domain), the ranking reverses on lines and narrows to a near-tie on words. On BanglaWriting, ground neither system trained on, **Swin** still wins word-task F1 (0.770 vs. 0.674). **The honest summary is dataset-conditional, not a single winner**: each system looks strongest on its own training domain, and **Swin** remains the stronger raw word detector on the one dataset built to be unbiased toward either. **BN-DRISHTI**'s edge on mask-mode mAP@75 is a rectangle-vs-polygon geometric artifact, independently confirmed via pixel-level analysis, not a real detection-quality gap.

References

[1] C. Wolf and J.-M. Jolion, "Object count/area graphs for the evaluation of object detection and segmentation algorithms," *IJDAR*, vol. 8, no. 4, pp. 280–296, 2006.

[2] A. Shahab, F. Shafait, and A. Dengel, "ICDAR 2011 Robust Reading Competition Challenge 2," *Proc. ICDAR*, 2011, pp. 1491–1496.

[3] D. Karatzas *et al.*, "ICDAR 2013 Robust Reading Competition," *Proc. ICDAR*, 2013, pp. 1484–1493.

[4] D. Karatzas *et al.*, "ICDAR 2015 Competition on Robust Reading," *Proc. ICDAR*, 2015, pp. 1156–1160.

[5] C. K. Ch'ng and C. S. Chan, "Total-Text," *Proc. ICDAR*, 2017, pp. 935–942.

[6] C. K. Ch'ng, C. S. Chan, and C.-L. Liu, "Total-Text: Toward Orientation Robustness," *IJDAR*, vol. 23, pp. 31–52, 2020.

[7] Y. Liu *et al.*, "Tightness-Aware Evaluation Protocol for Scene Text Detection," *Proc. CVPR*, 2019.

[8] Y. Baek *et al.*, "CLEval: Character-Level Evaluation for Text Detection and Recognition Tasks," *Proc. CVPRW*, 2020.

[9] T.-Y. Lin *et al.*, "Microsoft COCO," *Proc. ECCV*, LNCS vol. 8693, 2014, pp. 740–755.

[10] F. Simistira *et al.*, "ICDAR2017 Competition on Layout Analysis for Challenging Medieval Manuscripts," *Proc. ICDAR*, 2017.

[11] X. Zhong, J. Tang, and A. Jimeno Yepes, "PubLayNet," *Proc. ICDAR*, 2019, pp. 1015–1022.

[12] B. Pfitzmann *et al.*, "DocLayNet," *Proc. KDD*, 2022, pp. 3743–3751.

[13] G. Jaume, H. K. Ekenel, and J.-P. Thiran, "FUNSD," *Proc. ICDAR-OST*, 2019.

[14] Md. A. Rahman *et al.*, "BN-HTRd," *arXiv:2206.08977*, 2022.

[15] S. M. Jubaer *et al.*, "BN-DRISHTI," *ICDAR 2023 Workshops*, LNCS vol. 14193, Springer, 2023, pp. 186–202.

[16] M. F. Mridha *et al.*, "BanglaWriting: A Multi-Purpose Offline Bangla Handwriting Dataset," *Data in Brief*, vol. 34, 106633, 2021.

[17] Sk. Md. Obaidullah *et al.*, "PHDIndic_11," *Multimedia Tools and Applications*, vol. 77, pp. 1643–1678, 2018.

[18] Z. Liu *et al.*, "Swin Transformer," *Proc. ICCV*, 2021, pp. 10012–10022.

[19] K. He *et al.*, "Mask R-CNN," *Proc. ICCV*, 2017, pp. 2961–2969.

[20] N. Otsu, "A Threshold Selection Method from Gray-Level Histograms," *IEEE Trans. SMC*, vol. 9, no. 1, pp. 62–66, 1979.